

Douglas Ronne

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INTRODUCTION

I am collaborative, driven by the success of my team, take responsibility, and have an adaptive-flexible approach to problem solving. In addition to being very technically capable I am a skilled communicator and easy to get along with. I am driven to solve problems for the people around me whether my colleagues or customers. I love embedded software, motor control, sensors and measurement, and productivity. I am a citizen of both the U.S.A. and Canada.

EXPERIENCE

Embedded Software Engineer - Apex Amazon, Seattle, WA, United States

February 2025 - August 2025

- ◇ Found and corrected deficiencies in control scheme of prior BLDC motor control software leading to high current use
- ◇ Worked with team to design new board for a specific application
- ◇ Created motor control software for initial prototype that uses 30% the power as previous version and is quieter
- ◇ Also helped broader team work through some difficult problems. I do not believe my responsibility to a project is restricted to my specialty

R&D Manager - Protocase Inc, Sydney, NS, CANADA

September 2009 - January 2024

- ◇ As the sole developer I created from scratch our flagship sheet metal enclosure CAD software on schedule. This software brings in about \$5mil in revenue and about 20% of new customers.
- ◇ I grew the software team from 1 person to 10 people allowing us to expand our reach and not just maintain/ grow our CAD software but also create custom ERP software and much more.
- ◇ I took over the mechanical and industrial engineering teams and expanded the team from myself as the sole contributor to a high-performing team of about 25 professionals.
- ◇ Created custom formula parser language for dynamic parsing of variables and sub formulas in template CAD models.
- ◇ Visited customers to help them use our tools and better understand what they needed from us.
- ◇ Developed internal Automated Work Instructions on top of CAD to go from customer design to shop build sheets reducing rework by as much as 30% at some processes.
- ◇ Actively contributed to executive management, providing strategic direction and problem solving, helping to grow the company from 40 people and \$1mil in revenue to 400 people and over \$50mil in revenue.
- ◇ Created vision software to optimize print bed part location. Reduced rework rate by 50% and doubled productivity.

Head of Software - Power Control Engineering, Grants Pass, OR

May 2005 - May 2009

- ◇ As the sole developer I created embedded software for a Separately Excited Motor Controller for high-powered golf carts.
- ◇ Created a custom build and debug system in Python that shortened deploy time by about 50%.
- ◇ Discovered and reported a bug in the Texas Instruments C Compiler involving large switch statements.

Senior Multidisciplined Engineer 3 - Raytheon, El Segundo, CA

August 2001 - May 2005

- ◇ Designed algorithms for anti-jam GPS and radar systems, as well as conducted lab testing.
- ◇ Delivered 3 projects on time and under budget as the team lead of programmers and analyst.
- ◇ Utilized Matlab, C, C++, Ada, VHDL, and microcode on FPGAs for large matrix inversion.
- ◇ Demonstrated solutions and updates live to customers.

Math Teacher - Western Washington University, Bellingham, WA

August 1997 - June 2001

- ◇ Taught algebra, precalculus, and business calculus classes.
- ◇ Operated a math tutoring lab and tutored everything from algebra to multivariate calculus.

SKILLS

- ◇ **Languages:** Assembly, Visual Basic, C/C++/C#, Groovy, Java, Kotlin, LaTeX, Matlab, Python, Rust, Scala
- ◇ **Programming:** Agile Methods, Test Driven Development, Behavior Driven Development, Automated Testing, Refactoring, Design Patterns, Legacy Code Techniques, GitHub, GitLab, Continuous Integration, Continuous Delivery, Jenkins, gitlab-ci, CodeScene, Threading, Microservices, IntelliJ, Netbeans, Emacs, Linux, Windows, VMWare, 3D Modeling, Data Analysis, System Analysis, Risk Assessment, Debugging, GDB, Linkers, Makefiles, Code Review, Scripting, Bash, Distributed Systems, DevOps, Customer Feedback

MOST INFLUENTIAL BOOKS IN MY BOOKSHELF

“Test Driven Development”, “Growing Object Oriented Software, Guided By Tests”, “Refactoring”, “Refactoring to Patterns”, “Working Effectively with Legacy Code”, “Your Code as a Crime Scene”, “Tidy First”, “Clean Code”, “Crucial Conversations”, “Getting Things Done - The Art of Stress-free Productivity”, “Extreme Ownership”

EDUCATION

Western Washington University, Bellingham, WA 2001
Mathematics, MS - 4.0 GPA
Courses in numerical computing, windows programming, theoretical and applied math.

Western Washington University, Bellingham, WA 1999
Mathematics, BS - 3.96 GPA
Courses in numerical computing, data structures, assembly language.

South Puget Sound Community College, Olympia, WA 1996
General Studies, Associate of Arts - 3.92 GPA
Courses in math, computer science, english, physical and social sciences.

South Puget Sound Community College, Olympia, WA June 1993
Electronics, Associate of Technical Arts - 3.88 GPA
Courses in theory and practical electronics, machine language, assembly programming, and troubleshooting.

PROFESSIONAL DEVELOPMENT

Building an AI Application, Goto Masterclass 2024

Goto Conference, Chicago 2024

Front end web development in Rust, Udemy 2024

RustConf 2024 2024

Build a RC5 Symmetric Block Cipher in Rust, Udemy 2024

Ultimate Rust Crash Course, Udemy 2024

Ultimate Rust 2: Intermediate Concepts Udemy, 2024

Learn Rust by Building Real Applications Udemy, 2024

The World's Best Intro to TDD, jbrains.ca 2024

Level 1 Fire Training, Nova Scotia Fire School 2020

Protocase Conflict Management, Protocase 2018

SolidWorks API Training, SolidWorks 2014

Power Control Engineering Workshop, University of Wisconsin 2007

Raytheon Six Sigma, Raytheon 2003

Raytheon Systems Engineering, Raytheon 2004

Multirate Signal Processing, fred harris 2002

VOLUNTEERING

Lieutenant, Baddeck Volunteer Fire Department, Baddeck, NS, CANADA

September 2012 - June 2022

I started as a rank and file firefighter and became the lieutenant on our rescue truck and the Treasurer. Also took level 1 firefighter training.

Generalist, Pan Cape Breton Food Hub, Baddeck, NS, CANADA

September 2014 - June 2022

I helped the executive director design processes and assisted with general operations, maintenance, and labor.

Volunteer and Board Chair, Baddeck Farmers Market, Baddeck, NS, CANADA

March 2009 - June 2014

I helped the executive director design processes and assisted with general operations, maintenance, and labor. I was the board chair for 2 years.